Python for AI & Machine Learning

Class 25 – Data Visualization with Matplotlib

Course: Python for AI & Machine Learning
Module: Data Visualization using Python
Duration: 2 Hours (Theory + Practical)

Learning Objectives

By the end of this session, students will be able to:

- Understand the importance of Data Visualization.
- Install and import the Matplotlib library.
- Create different types of charts.
- Customize graphs with titles, labels, legends, and colors.
- Save charts as image files.
- Apply visualization techniques in AI & Machine Learning projects.

Revision

In the previous class, we learned:

- Data Aggregation
- `groupby()`
- Aggregate Functions
- Pivot Tables
- `value_counts()`
- `unique()`
- Multi-level Grouping

Today we will learn how to represent data visually using **Matplotlib**.

What is Data Visualization?

Data Visualization is the graphical representation of data using charts and graphs.

Instead of reading tables containing hundreds of numbers, charts help us identify:

- Trends
- Patterns
- Comparisons

- Relationships
- Outliers

Visualization makes data easier to understand and analyze.

Why Do We Need Data Visualization?

Imagine a company has sales data for 12 months.

Looking at a table of numbers makes it difficult to identify growth.

A line chart immediately shows:

- Increasing sales
- Decreasing sales
- Seasonal trends
- Peak months

Visualization helps in making faster and better decisions.

What is Matplotlib?

Matplotlib is one of the most popular Python libraries used for creating graphs and charts.

It is widely used in:

- Data Science
- Artificial Intelligence
- Machine Learning
- Business Analytics
- Scientific Research
- Financial Analysis

Installing Matplotlib

Install using pip:

```
pip install matplotlib
```

Verify installation:

```
pip show matplotlib
```

Importing Matplotlib

```
import matplotlib.pyplot as plt
```

`pyplot` provides functions to create charts.

Creating a Simple Line Chart

```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May"]

sales = [25,40,35,50,60]

plt.plot(months, sales)

plt.show()
```

Adding Title and Labels

```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May"]
sales = [25,40,35,50,60]

plt.plot(months, sales)

plt.title("Monthly Sales")

plt.xlabel("Months")

plt.ylabel("Sales")

plt.show()
```

Changing Line Style

```
plt.plot(
    months,
    sales,
    color="blue",
    linestyle="--",
    marker="o"
)

plt.show()
```

Bar Chart

Used to compare categories.

```
import matplotlib.pyplot as plt

courses = ["Python","AI","ML","Cloud"]

students = [45,35,30,25]

plt.bar(courses, students)

plt.title("Students Enrolled")

plt.show()
```

Horizontal Bar Chart

```
plt.barh(courses, students)

plt.show()
```

Pie Chart

Used to show percentages.

```python
import matplotlib.pyplot as plt

courses = ["Python","AI","ML"]

students = [45,35,20]

plt.pie(
    students,
    labels=courses,
    autopct="%1.1f%%"
)

plt.show()
```

Histogram

Used to display frequency distribution.

```python
import matplotlib.pyplot as plt

marks = [65,70,75,80,85,90,95,80,75,70]

plt.hist(marks)

plt.show()
```

Applications:

- Student performance
- Salary distribution
- Age analysis

Scatter Plot

Used to identify relationships between two variables.

```python
import matplotlib.pyplot as plt

hours = [1,2,3,4,5]

marks = [45,55,65,80,90]

plt.scatter(hours, marks)
```

```
plt.show()
```

Applications:

- Study hours vs marks
- Experience vs salary
- Advertising cost vs sales

Adding Grid

```
plt.plot(months, sales)

plt.grid(True)

plt.show()
```

Adding Legend

```
plt.plot(
    months,
    sales,
    label="Sales"
)

plt.legend()

plt.show()
```

Saving a Chart

```
plt.savefig("sales_chart.png")
```

The graph will be saved as an image.

Multiple Line Charts

```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr"]

sales_2025 = [20,30,40,50]
sales_2026 = [25,35,45,60]

plt.plot(months, sales_2025, label="2025")

plt.plot(months, sales_2026, label="2026")

plt.legend()

plt.show()
```

AI Example 1 – Student Performance

```
import matplotlib.pyplot as plt

students = ["Rahul","Sneha","Aman","Priya"]

marks = [82,95,76,91]

plt.bar(students, marks)

plt.title("Student Marks")

plt.show()
```

AI Example 2 – Machine Learning Accuracy

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5]

accuracy = [65,72,80,88,94]

plt.plot(epochs, accuracy, marker="o")

plt.title("Model Accuracy")
```

```
plt.xlabel("Epoch")

plt.ylabel("Accuracy (%)")

plt.show()
```

AI Example 3 – Sales Dashboard

```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr"]

sales = [25000,30000,45000,50000]

plt.bar(months, sales)

plt.title("Monthly Revenue")

plt.show()
```

AI Example 4 – Age Distribution

```
import matplotlib.pyplot as plt

ages = [18,20,21,22,23,21,20,24,25,22]

plt.hist(ages)

plt.title("Student Age Distribution")

plt.show()
```

Advantages of Matplotlib

- Easy to use
- Supports multiple chart types
- Highly customizable
- Integrates with NumPy and Pandas
- Widely used in AI and Data Science
- Produces publication-quality graphs

Best Practices

- Always give charts a meaningful title.
- Label the X-axis and Y-axis.
- Use legends when comparing multiple datasets.
- Choose the correct chart type.
- Keep graphs simple and readable.
- Save important charts for reports and presentations.

Practical Exercise 1

Create a line chart showing monthly sales.

Practical Exercise 2

Create a bar chart showing student enrollment in different courses.

Practical Exercise 3

Create a pie chart showing the percentage of students in Python, AI, and ML courses.

Practical Exercise 4

Create a histogram showing the distribution of student marks.

Practical Exercise 5

Create a scatter plot showing study hours versus marks.

Assignment

AI Sales Dashboard

Create a Data Visualization project that:

- Reads sales data from a CSV file.

- Displays:
- Line Chart
- Bar Chart
- Pie Chart
- Histogram
- Scatter Plot
- Add titles, axis labels, legends, and grid.
- Save all charts as PNG image files.

Bonus Challenge:

Create one dashboard comparing sales for two different years using multiple line charts.

Interview Questions

1. What is Data Visualization?
2. Why is Data Visualization important?
3. What is Matplotlib?
4. What is the purpose of `pyplot`?
5. What is the difference between a Line Chart and a Bar Chart?
6. When should you use a Pie Chart?
7. What is a Histogram used for?
8. What is a Scatter Plot?
9. How do you save a chart in Matplotlib?
10. How is Matplotlib used in AI and Machine Learning?

Summary

Today we learned:

- Introduction to Data Visualization
- Installing Matplotlib
- Importing `matplotlib.pyplot`
- Line Charts
- Bar Charts
- Pie Charts
- Histograms
- Scatter Plots
- Titles, Labels, Legends, and Grid
- Saving Charts
- AI & Machine Learning Applications

Data Visualization is a key skill for every Data Scientist and Machine Learning Engineer. It helps transform raw data into meaningful insights, making analysis and decision-making easier.

Next Class Preview

Class 26 – Advanced Data Visualization with Matplotlib & Seaborn

Topics to be covered:

- Figure Size and Style
- Subplots
- Box Plots
- Violin Plots
- Heatmaps
- Pair Plots
- Count Plots
- Correlation Matrix
- Introduction to Seaborn
- Data Visualization in AI & Machine Learning Projects